

Sreedev TS

AI/ML Backend Engineer · GenAI & Agentic Systems

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SUMMARY

AI/ML backend engineer with 2+ years solving real infrastructure problems in hospitals — building production GenAI systems deployed on-premise under GDPR, with no managed cloud services. **3× national hackathon finalist.**

PROJECTS & HACKATHONS

MIT Summer School — AI + Physics

Aug 2026

NSF IAIFI · UMass Boston

Virtual

- Week-long NSF institute on machine learning for the physical sciences: symbolic regression, diffusion and generative modelling, computer vision, and simulation-based inference; lectures from Miles Cranmer (author of PySR, Cambridge), Jun-Yan Zhu (CMU), and Christoph Weniger (Amsterdam).
- Ran equation discovery on real datasets with allocated **HPC** compute — multi-population evolutionary search in PySR and SymbolicRegression.jl, custom operators and loss constraints, and Pareto selection trading accuracy against expression complexity.

AWS × Elastic Hackathon — Finalist

Sep 2026

Finalist from 8,500+ registrations · PROBE · Bengaluru

hackculture.io/hackathons/forge-the-future-hackathon-2026

- Production incidents need an engineer to read dashboards and find root cause: PROBE detects the anomaly, diagnoses it, remediates, and turns each developer fix into a runbook entry so the next occurrence is handled automatically.
- **Stack:** Elastic Agent Builder and Workflows, Elasticsearch anomaly detection, Amazon Bedrock (Claude), Jira, Kibana cases.

iQOO City Battles Hackathon — Finalist

Finalist from 13,000+ participants · MUDRA+ · Chennai

pisquarelabs.github.io/mudra

- Signing a full sentence is slow and error-prone on a phone call: MUDRA+ autocompletes 2–3 key ASL signs into a spoken sentence, running entirely on-device with no cloud APIs.
- **Stack:** React Native, MediaPipe hand landmarks, on-device Qwen2.5-3B (Q4_K_M) via llama.cpp, Whisper, SQLite, Android TTS.

Razorpay × Sarvam AI Buildathon

Jul 2026

Finalist from 3,000+ applicants · NyayBandh · Bengaluru

built.growthx.club/projects/nyaybandh

- Negotiations fail over misunderstandings nobody catches in the moment: NyayBandh mediates the conversation live, flagging where the two parties diverge before it becomes a dispute.
- **Stack:** Sarvam AI models, LangGraph orchestration, Python; shipped end-to-end within the 8-hour window.

Hacklogics — Winner

Winner from 15 teams · Board Flex · Thiruvananthapuram

Simplogics internal

- College events were scheduled and scored on spreadsheets: Board Flex handles scheduling, event management, and live score tracking in one place, built in 16 hours.
- **Stack:** Python, FastAPI, PostgreSQL, OpenAI API, AWS EC2.

EXPERIENCE

Software Engineer (AI/ML Backend) — Simplogics

Jul 2024 – Present

MOM: hospital BI platform deployed on-premise in customer environments

Technopark, Thiruvananthapuram

Agentic Analytics Dashboard

- Built and deployed a production MCP server exposing **100+** internal APIs as typed tools for agents.
- Developed a LangGraph ReAct agent over **10K+** medico-economic records, producing drill-down cost reports for analysts.
- Cut LLM inference cost by up to **80%** with context engineering: Redis memory, vector-based tool selection, batching, model routing, and prompt caching.
- Built a conversational analytics UI that renders interactive charts inline from agent output.

NLP-to-SQL Chatbot

- Built a Rasa NLP-to-SQL engine serving **150K+** production queries, ranking 100+ SQL templates by intent, entity, and vector similarity.
- Added a self-learning loop that generates training data and spelling corrections from live queries, improving query accuracy **36%**.
- Built a multi-pipeline NLP framework with automated precision/recall monitoring and versioned model benchmarks.

Health Pathway Analysis

- Built a GDPR-compliant pipeline cleaning **200K+** noisy patient records to reconstruct cohort treatment pathways.
- Modelled patient journeys as graphs, using BFS/DFS to compute pathway-level treatment costs.
- Ran Monte Carlo and Erlang-C queuing simulations over **10,000+** scenarios to find staffing bottlenecks.

Platform & Infrastructure

- Built PHI pseudonymisation that keeps identifiers out of LLM context under GDPR.
- Designed an event-driven Redis architecture moving reports, ETL, and inference to async workers.
- Set up a Grok bot for building demo POCs quickly on top of production apps.
- Ran the stack fully self-hosted: Python and Node.js services on MSSQL, Jenkins CI/CD with Nexus artifact mirroring, and Grafana, Prometheus, and Loki for observability.

EDUCATION

B.Tech, Computer Science Engineering — CGPA 7.71
Marian Engineering College, APJ Abdul Kalam Technological University

Dec 2020 – Jun 2024
Thiruvananthapuram

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript (Node.js), Java, C
AI & GenAI	MCP, LangGraph, ReAct agents, RAG, context engineering, model routing, prompt caching, ChromaDB, Amazon Bedrock, llama.cpp, MediaPipe
NLP	Rasa, spaCy, NER, intent and entity extraction, embeddings, precision/recall evaluation
Backend & Data	FastAPI, Node.js, Redis, event-driven architecture, MSSQL, PostgreSQL, Elasticsearch, ETL, Pandas
Infra & DevOps	On-premise deployment, Jenkins, Nexus, Grafana, Prometheus, Loki, AWS (EC2, Bedrock), Git
Tools	Claude Code, Cursor, Grok bot